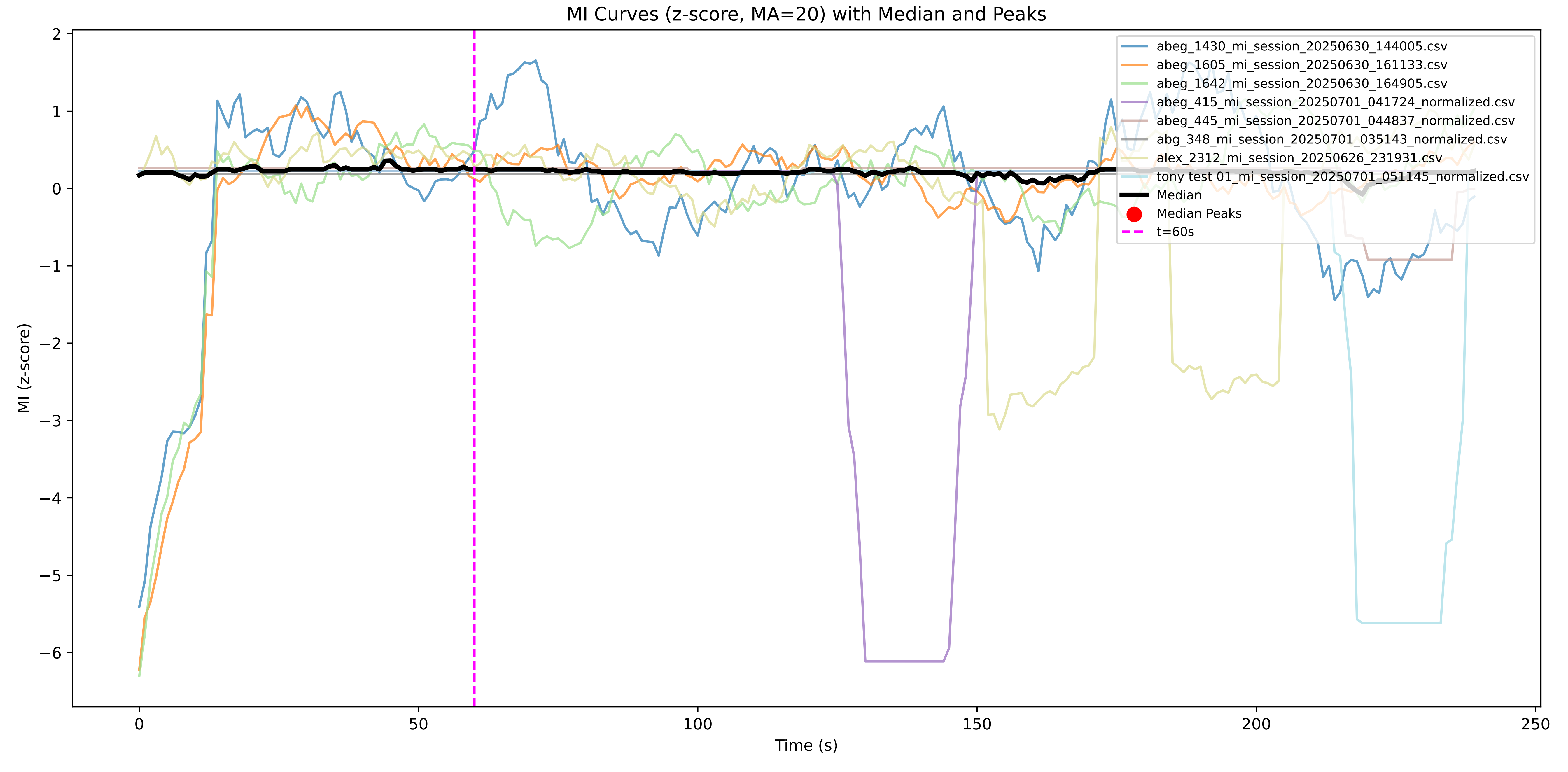
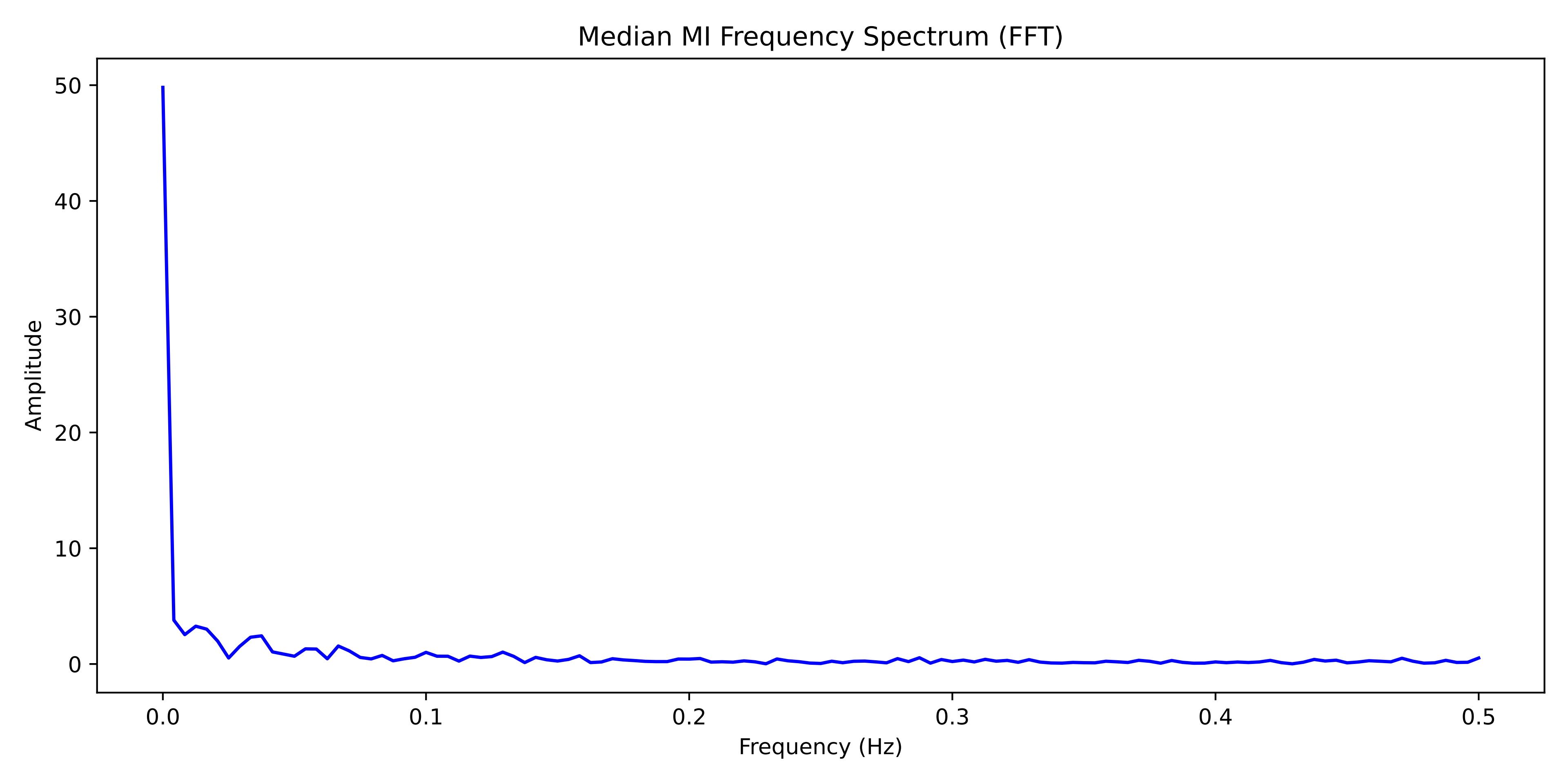
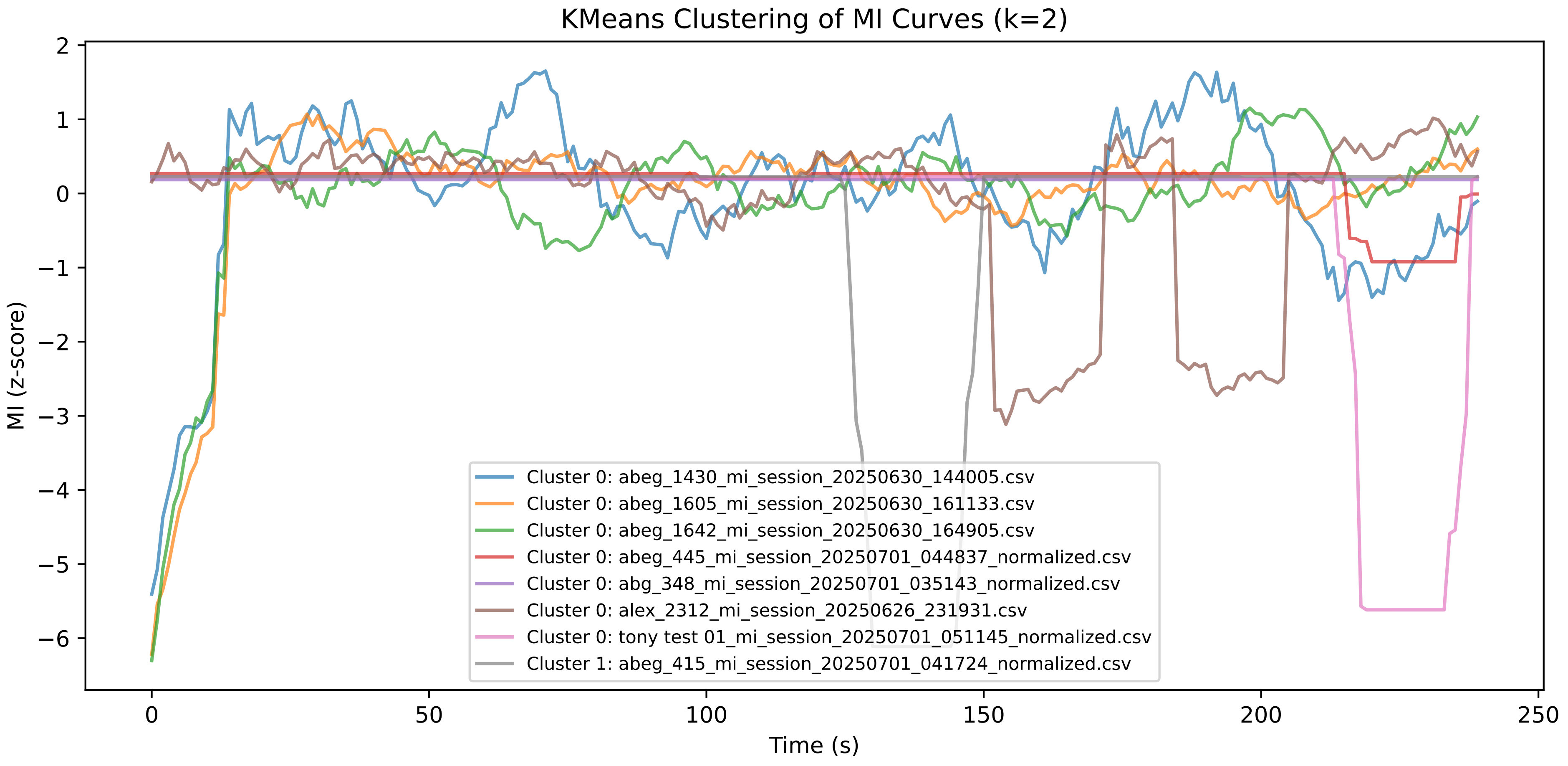
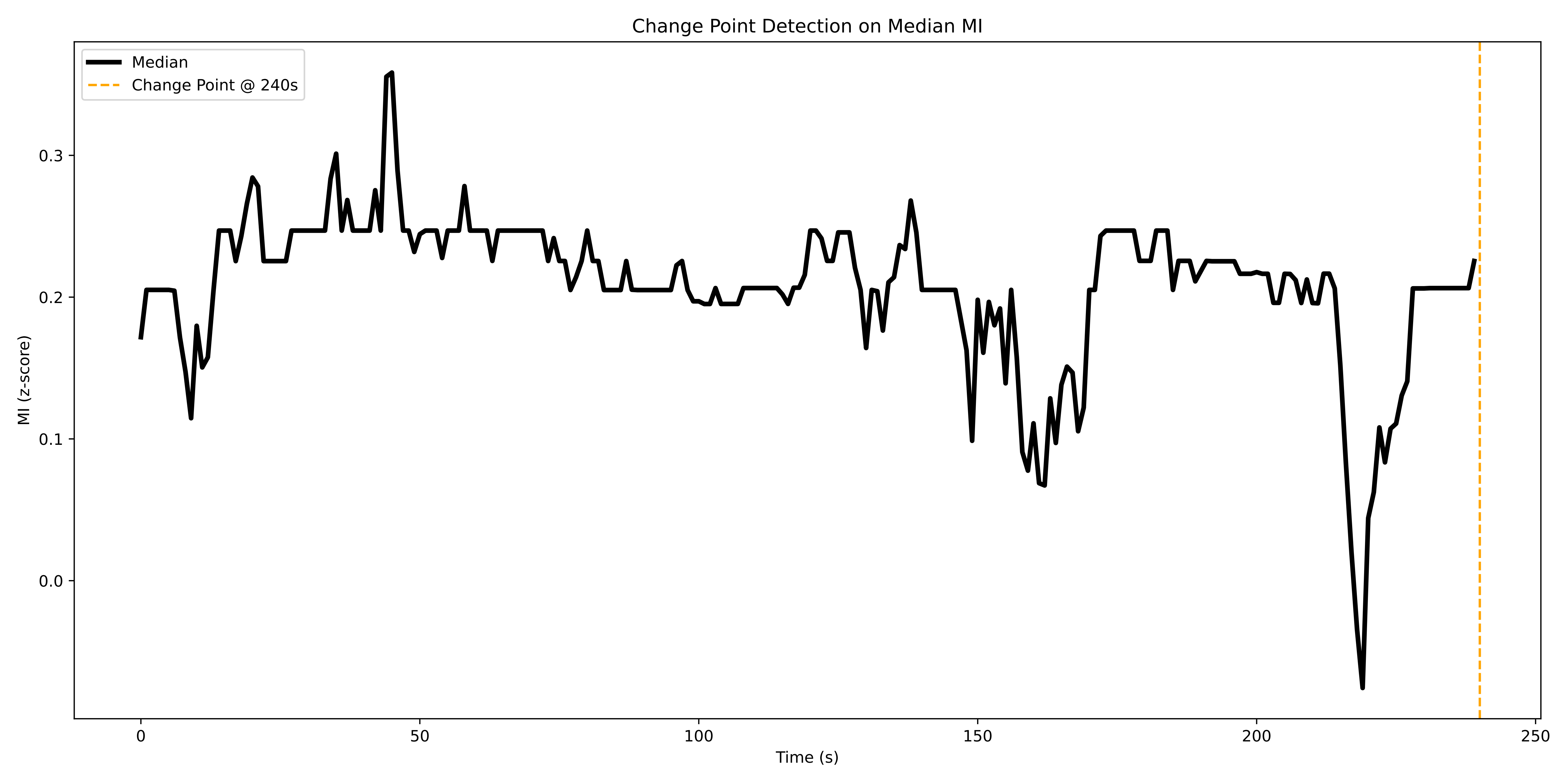


MI Curves (z-score, MA=20) with Median and Peaks
abeg_1430_mi_session_20250630_144005.csv
abeg_1605_mi_session_20250630_161133.csv
abeg_1642_mi_session_20250630_164905.csv
abeg_415_mi_session_20250701_041724_normalized.csv
abeg_445_mi_session_20250701_044837_normalized.csv
abg_348_mi_session_20250701_035143_normalized.csv
alex_2312_mi_session_20250626_231931.csv
tony test 01_mi_session_20250701_051145_normalized.csv
Median
Median Peaks
t=60s
MI (z-score)
Time (s)
2
1
0
−1
−2
−3
−4
−5
−6
0
50
100
150
200
250

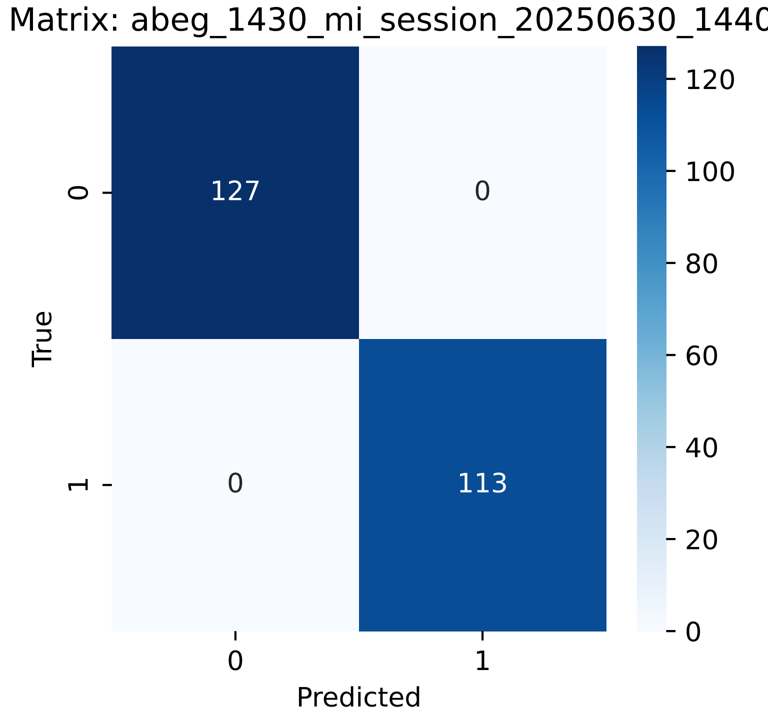


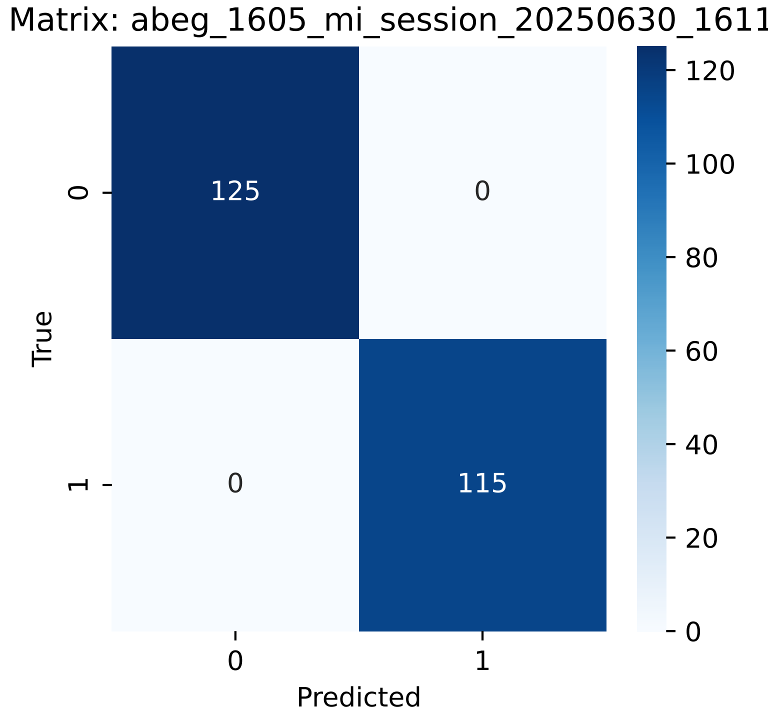
Median MI Frequency Spectrum (FFT)
Amplitude
Frequency (Hz)
0
10
20
30
40
50
0.0
0.1
0.2
0.3
0.4
0.5


KMeans Clustering of MI Curves (k=2)
MI (z-score)
Time (s)
2
1
0
−1
−2
−3
−4
−5
−6
0
50
100
150
200
250
Cluster 0: abeg_1430_mi_session_20250630_144005.csv
Cluster 0: abeg_1605_mi_session_20250630_161133.csv
Cluster 0: abeg_1642_mi_session_20250630_164905.csv
Cluster 0: abeg_445_mi_session_20250701_044837_normalized.csv
Cluster 0: abg_348_mi_session_20250701_035143_normalized.csv
Cluster 0: alex_2312_mi_session_20250626_231931.csv
Cluster 0: tony test 01_mi_session_20250701_051145_normalized.csv
Cluster 1: abeg_415_mi_session_20250701_041724_normalized.csv



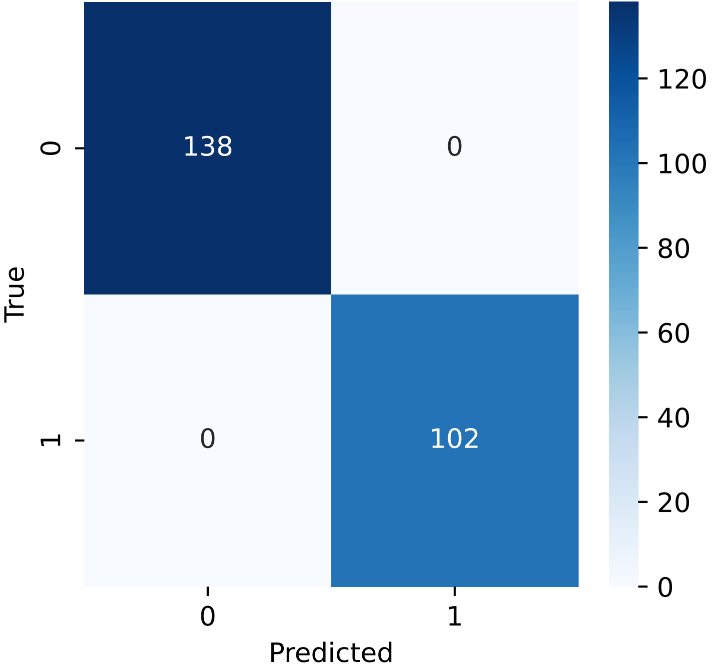
Change Point Detection on Median MI
Median
Change Point @ 240s
MI (z-score)
Time (s)
0.0
0.1
0.2
0.3
0
50
100
150
200
250

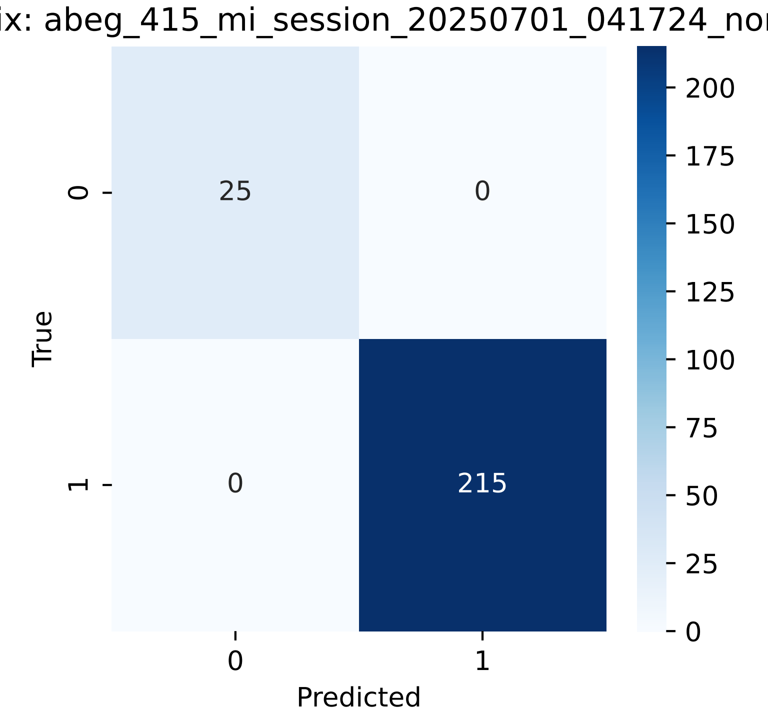

Matrix: abeg_1430_mi_session_20250630_1440
127
0
0
113
True
Predicted
0
1
0
1
120
100
80
60
40
20
0

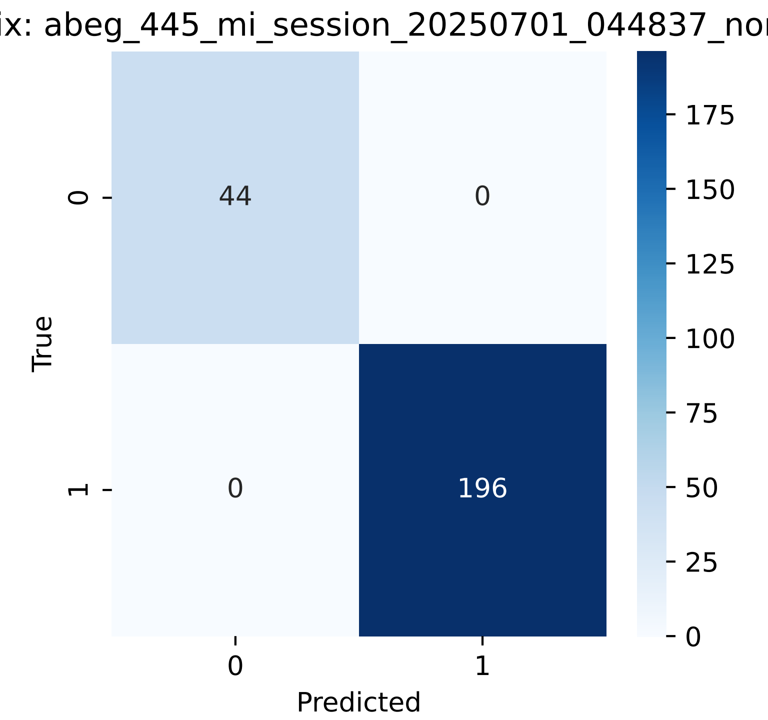


Matrix: abeg_1605_mi_session_20250630_1611
125
0
0
115
True
Predicted
0
1
0
1
120
100
80
60
40
20
0

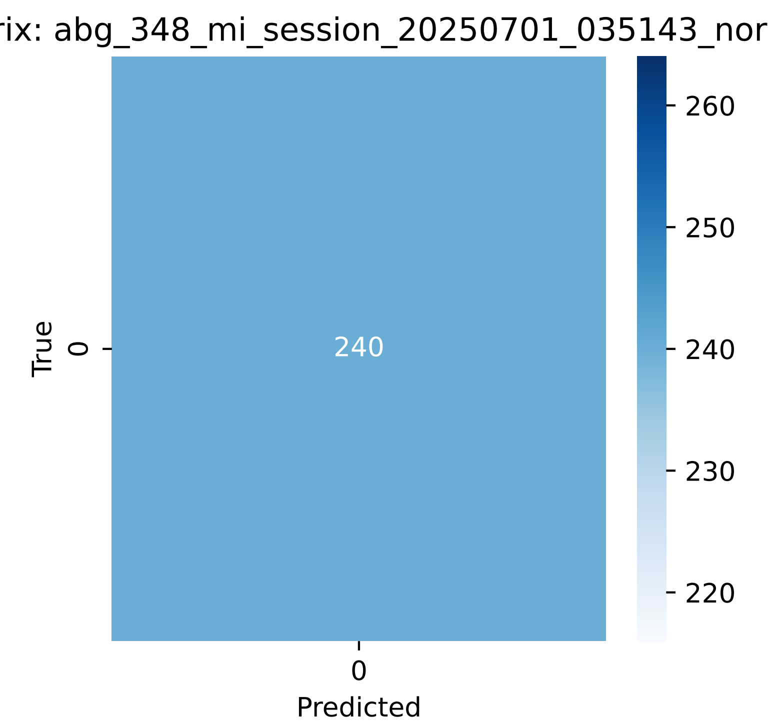
Matrix: abeg_1642_mi_session_20250630_1649

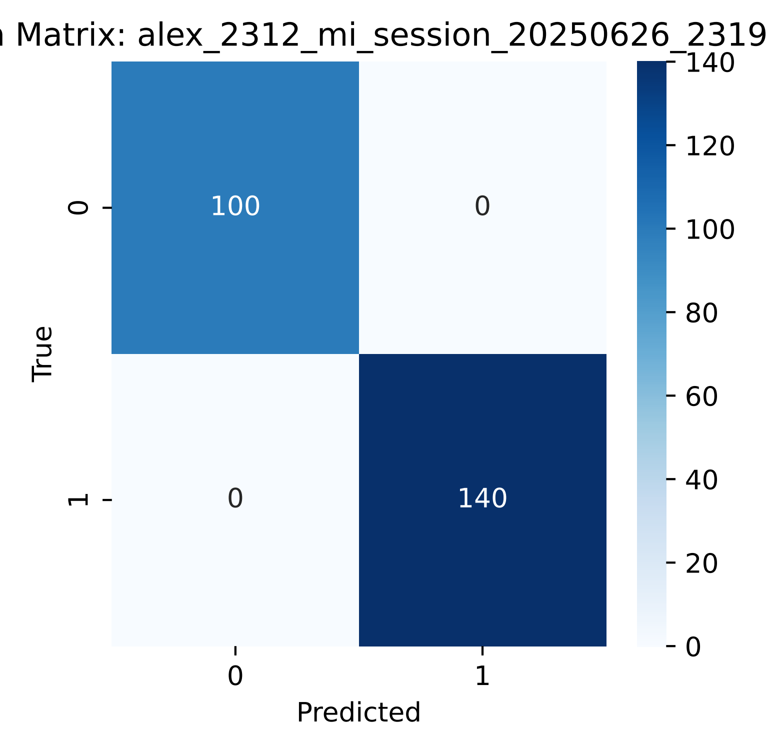


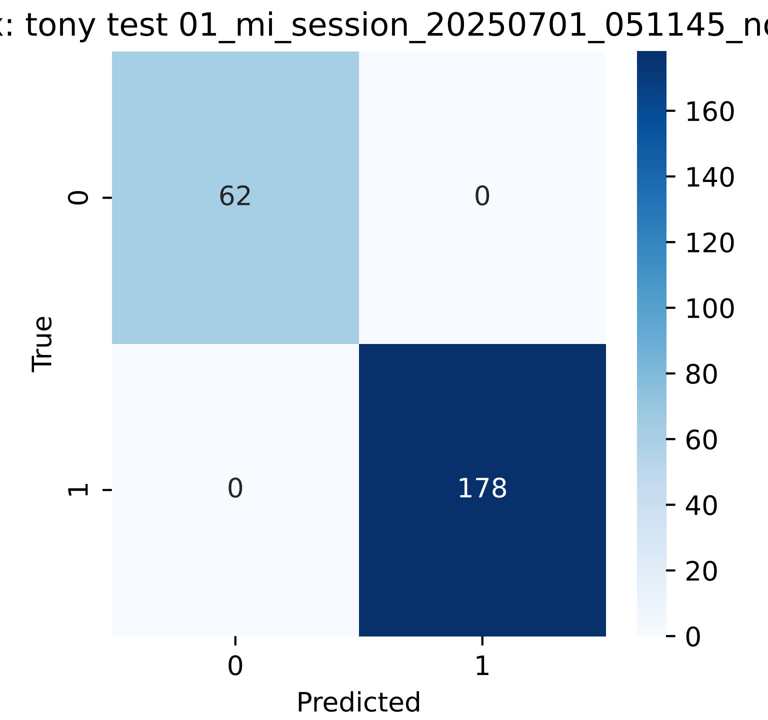

ix: abeg_415_mi_session_20250701_041724_no
True
Predicted
0
1
25
0
0
215
200
175
150
125
100
75
50
25
0

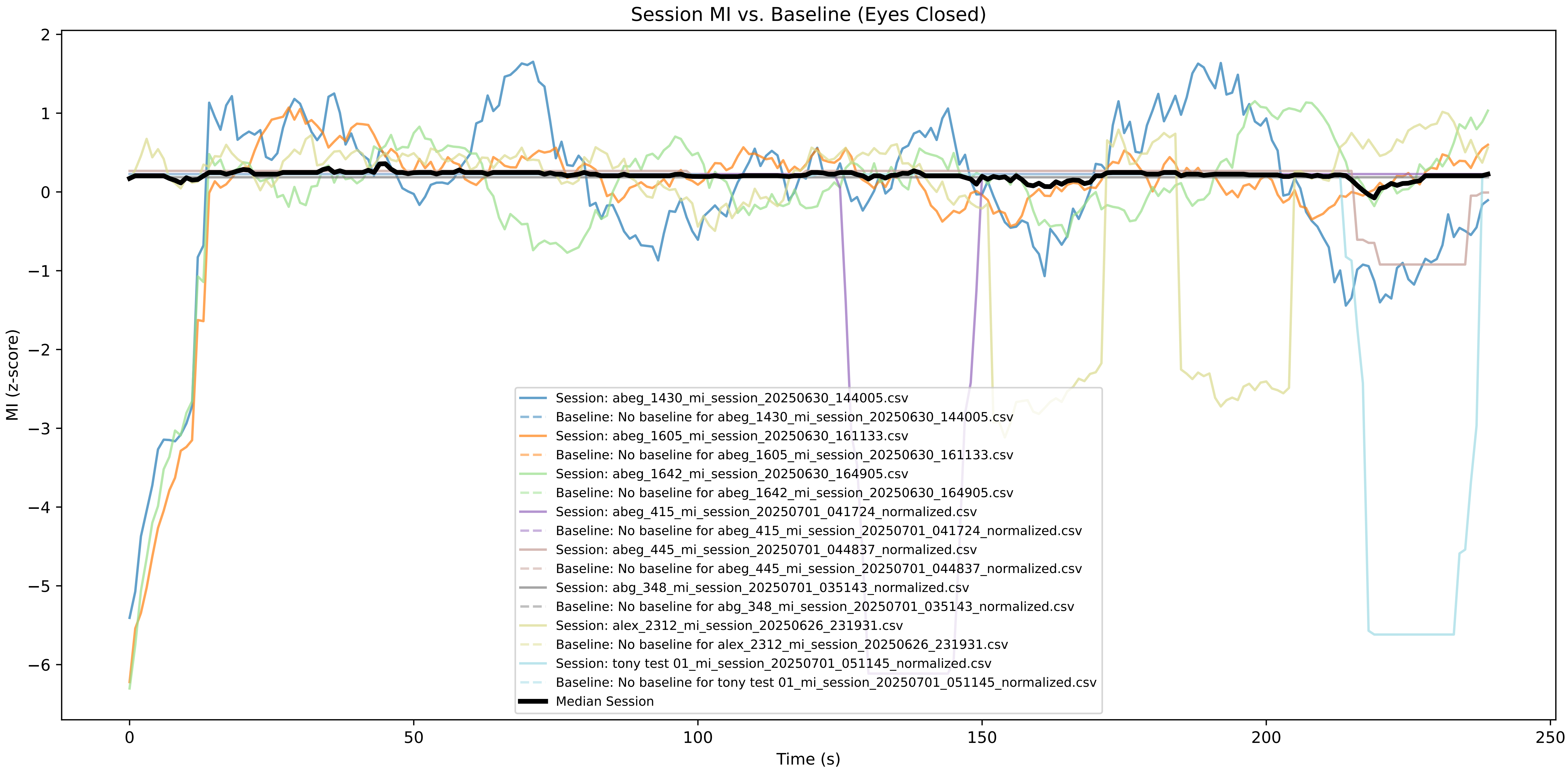


ix: abeg_445_mi_session_20250701_044837_no
44
0
0
196
True
0
1
Predicted
0
1
175
150
125
100
75
50
25
0


rix: abg_348_mi_session_20250701_035143_nor
True
0
240
0
Predicted
260
250
240
230
220


n Matrix: alex_2312_mi_session_20250626_2319
100
0
0
140
True
Predicted
0
1
0
1
140
120
100
80
60
40
20
0


x: tony test 01_mi_session_20250701_051145_no
True
Predicted
62
0
0
178
0
1
0
1
160
140
120
100
80
60
40
20
0


Session MI vs. Baseline (Eyes Closed)
MI (z-score)
Time (s)
2
1
0
−1
−2
−3
−4
−5
−6
0
50
100
150
200
250
Session: abeg_1430_mi_session_20250630_144005.csv
Baseline: No baseline for abeg_1430_mi_session_20250630_144005.csv
Session: abeg_1605_mi_session_20250630_161133.csv
Baseline: No baseline for abeg_1605_mi_session_20250630_161133.csv
Session: abeg_1642_mi_session_20250630_164905.csv
Baseline: No baseline for abeg_1642_mi_session_20250630_164905.csv
Session: abeg_415_mi_session_20250701_041724_normalized.csv
Baseline: No baseline for abeg_415_mi_session_20250701_041724_normalized.csv
Session: abeg_445_mi_session_20250701_044837_normalized.csv
Baseline: No baseline for abeg_445_mi_session_20250701_044837_normalized.csv
Session: abg_348_mi_session_20250701_035143_normalized.csv
Baseline: No baseline for abg_348_mi_session_20250701_035143_normalized.csv
Session: alex_2312_mi_session_20250626_231931.csv
Baseline: No baseline for alex_2312_mi_session_20250626_231931.csv
Session: tony test 01_mi_session_20250701_051145_normalized.csv
Baseline: No baseline for tony test 01_mi_session_20250701_051145_normalized.csv
Median Session

MI Advanced Report
Generated: 20250716_234946

Report Interpretation and Methodological Documentation

1. Mindfulness Index (MI) Definition
The Mindfulness Index (MI) is computed at 1 Hz from EEG/EDA data. MI is derived as a composite score re

2. Z-Scoring Method
All MI time series are z-score normalized using the mean and standard deviation of the entire session.

3. K-Means Clustering
K-means clustering (k=2) is applied to the set of MI curves, segmenting them into two groups representin

4. Event Markers
Key experimental time points are annotated on all plots:
30s, 90s, 150s, 240s, 300s (feedback cues, guided imagery, transitions).

5. Peak Analysis
Peaks are defined as local maxima in the median MI curve exceeding a z-score threshold of 1.5. Peak amp

6. Area Under Curve (AUC)
AUC is calculated for each MI curve and the median using the trapezoidal rule over the full session and sp

7. Bootstrap Confidence Interval
Bootstrap resampling (n=1000, 95% CI) estimates the confidence interval for the difference in mean MI b

8. Change Point Detection (Ruptures)
Change points in the median MI curve are identified using the Pelt algorithm with an L2 cost function. Pen

9. Time-Frequency Analysis (FFT)
The frequency spectrum of the median MI curve is computed using FFT with a rectangular window. Freque

10. Mixed-Effects Modeling (Statsmodels)
A linear mixed-effects model is fit: MI ~ time + (1|Session_File), accounting for repeated measures and in

11. ANOVA (Across Files/Subjects)
One-way ANOVA tests for differences in mean MI across sessions (average MI over the full session).

Descriptive Summary
- All MI session files with >= 240 samples were included.
- Each curve is smoothed (moving average window=20), z-scored, and plotted in a unique color.
- The median curve and detected peaks are highlighted.
- Event markers at 30, 90, 150, 240, and 300 seconds provide experimental context.
- Peak analysis, AUC, bootstrap CI, change point detection, FFT, clustering, mixed-effects modeling, and A
- Results show dynamic and heterogeneous patterns of mindfulness, with statistically significant peaks an

References:
[1] Tang et al., Nature Reviews Neuroscience, 16(4), 213–225.
[2] Lomas et al., Neuroscience & Biobehavioral Reviews, 57, 401–410.
[3] Cahn & Polich, Psychological Bulletin, 132(2), 180–211.

Statistical Results:
- Peak analysis: Median curve has 0 peaks (z > 1.5) at times [].
- AUC (median): 49.61
- Bootstrap 95% CI for mean diff (first vs. second half): [-0.01278779 0.01405814]
- Change points detected at: [240]
- FFT shows frequency content of median MI.
- KMeans clustering (k=2) groups curves by similarity.
- Mixed-effects model:

```
        Mixed Linear Model Regression Results
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```